Kalen Patterson

Software Engineer - AI Systems & CAD Automation
Indianapolis, IN | kcpatterson27@gmail.com | github.com/kcpatt27

Summary

Software engineer building constrained AI systems that act on physical design data. Built Fixture, a local-first harness that drives FreeCAD and Blender through a schema-validated tool contract, benchmarks planner models on a 60-case suite, and exports STEP/STL parts. Also published a local decision engine on PyPI and built a local-first LLM memory system; strongest where models, geometry APIs, and verification meet. Pursuing a B.S. in Computer Science at Western Governors University.

Skills

CAD and geometry: FreeCAD (Part/PartDesign, PySide6 workbench), Blender (`bpy`), scripted and parametric CAD, STEP/STL, glTF, headless geometry evaluation, mesh/STL workflows

AI and models: PyTorch, Ollama, OpenRouter, local inference (ONNX), JSON Schema, tool calling, AI-driven CAD generation, planner benchmarking, model calibration, evaluation and grounding, MCP

Robotics and simulation: PX4 SITL, Gazebo, MAVSDK, simulation-in-the-loop

Languages: Python, Rust, TypeScript, JavaScript, Go, SQL

Backend and web: FastAPI, Node.js, React, Next.js, PostgreSQL, SQLite, LanceDB, Prisma, Supabase, Cloudflare Workers

Practices: pytest, Docker, GitHub Actions, PyPI packaging, CLI tooling, Git

Projects

Fixture - local-first AI CAD harness

Python, FastAPI, FreeCAD, Blender, Ollama · github.com/kcpatt27/fixture-cad · Public / PyPI 0.2.0 · 2026

- Built a generative CAD system that turns chat into programmatic CAD operations in FreeCAD and Blender through one JSON-schema tool contract: named tools only, confirm-before-mutate, undo-wrapped changes, and no model-written code execution.
- Benchmarked 13+ local planners and 4 cloud models on a 60-case suite; Fixture passed 96.7–98.3% vs 57.6–74.6% for the closest open-source CAD assistant, with the failure analysis kept alongside the results.
- Gate PASS: pytest 140, FreeCAD 92/0, Blender 29/0. Built STEP/STL desk cases (4-piece laptop cradle, 2-piece under-desk shelf) and a Blender scan-cleanup tool that exports watertight STL; the keyboard case stays private.
- Reproduced a 1.2M-parameter next-action policy (0.98 eval success, 0.9996 mean IoU) and built a goal compiler that reached IoU 1.0000 on real desk parts.

AI Security Layer (AISL) - agent execution runtime

TypeScript, Node, Postgres, Ed25519 · Private / walkthrough available · 2026

- Built a capability-gated execution layer for AI agents where the model cannot grant itself authority: policy evaluation, taint tracking, egress DLP, and a signed hash-chained audit log. Documented as a spec: 39 security claims, 32 ADRs, 716 tests, and a 12-case adversarial corpus.

allSoft

Python, PX4 SITL, Gazebo, Docker, MAVSDK · robot simulation validation · 2026

- Built a YAML-driven simulation-in-the-loop harness: PX4 SITL + Gazebo in Docker, MAVSDK missions, JSON/SQLite artifacts for regression comparison, and a live/dry-run split; 4/4 reference scenarios pass live.

Living State Machine - local memory system

Python, FastAPI, Ollama, LanceDB · github.com/kcpatt27/living-state-machine · 2026

- Built a local-first LLM memory system with six ablatable memory mechanisms, reaching 1.000 vs 0.100 preference-recall accuracy at matched context budget; 240+ tests and a preprint in preparation.

Experience

Software Engineer, NextDish 2024 – 2026

Next.js, TypeScript, Supabase, Cloudflare, OpenRouter · Private

- Architected and solo-built a full-stack AI platform: 131+ React components, ~40 API endpoints, a 20+ table schema, and 339 commits over 14 months.
- Implemented hybrid AI generation: a client-side Llama 3.2 model (ONNX) for offline generation, plus an OpenRouter server tier with dynamic model discovery and automatic fallback.
- Refactored 50+ Supabase row-level-security policies across 20 tables and removed 12 unused indexes, clearing all performance-advisor warnings.

Education

Western Governors University, B.S. in Computer Science
Currently pursuing

Independent research, Living State Machine paper
preprint in preparation, 2026

Additional Work

Also published eyewright (PyPI) and mcp-weight (npm; 517 downloads in its first month).